TM Forum Specification

Resource Inventory Management User Guide

TMF639

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Introduction

The following document is intended to provide details of the REST API interface for the Resource Inventory. The intent of this API is to provide a consistent/standardized mechanism to query and manipulate the resources of the inventory.

It includes the model definition as well as all available operations.

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Sample Use Cases

Reader will find example of use cases using Usage API in IG1228 “ODA end-to-end use cases” document.

Resource inventory query for a party (Use Case 1)

The Resource Inventory API can be used to query the resource instances for a party playing the role of customer via Self Service Portal or the Call Centre operator can query the resource instances on behalf of the customer while a customer may have a complaint or a query.

Resource inventory create/update as part of resource provisioning (Use Case 2)

The Resource Inventory API can be called by the Resource Order Management to create a new resource instance/ update an existing resource instance in the Resource Inventory.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a PhysicalResource and LogicalResource inheriting properties from the abstract Resource entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Resource instances some may be instances of PhysicalResource where other could be instances of LogicalResource. The @type gives this information. All resources and sub-resources of this API have a @type attributes that must be provided.

The @referredType can be used within reference entities (like for instance an ResourceRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type denotes the class type of the reference itself, such as PhysicalResourceRef or LogicalResourceRef, and not the class type of the referred object.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

LogicalResource resource

Logic resource is a type of resource that describes the common set of attributes shared by all concrete logical resources (e.g. TPE, MSISDN, IP Addresses) in the inventory.

Resource model

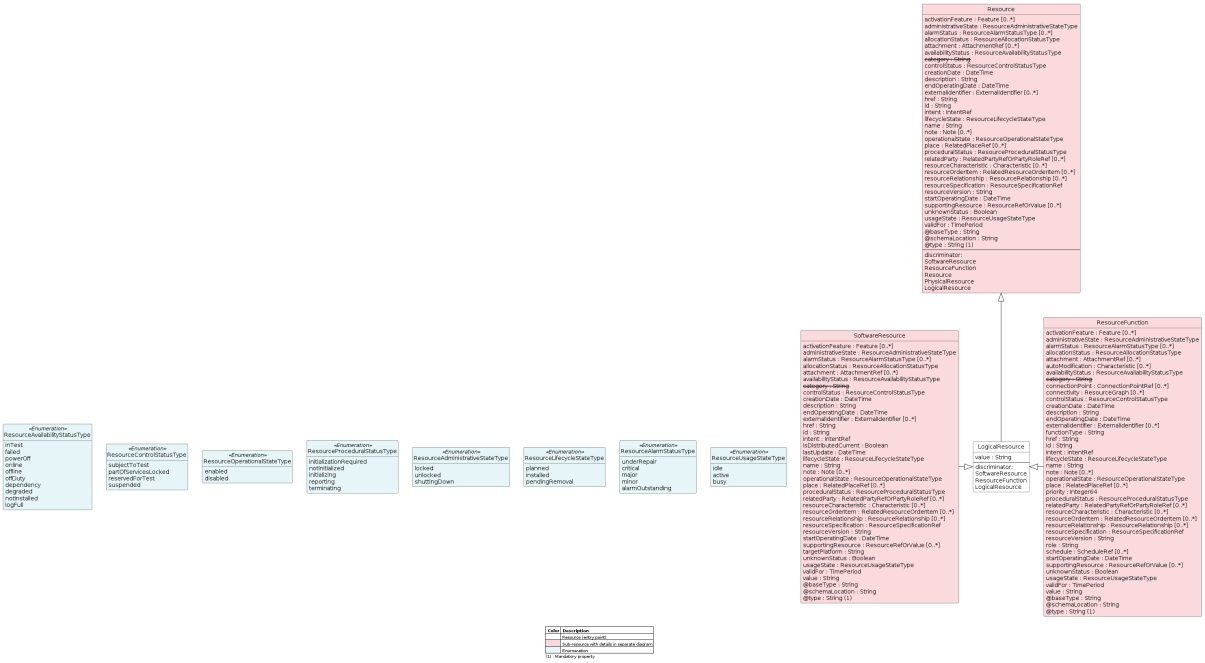


Figure 1 - LogicalResource

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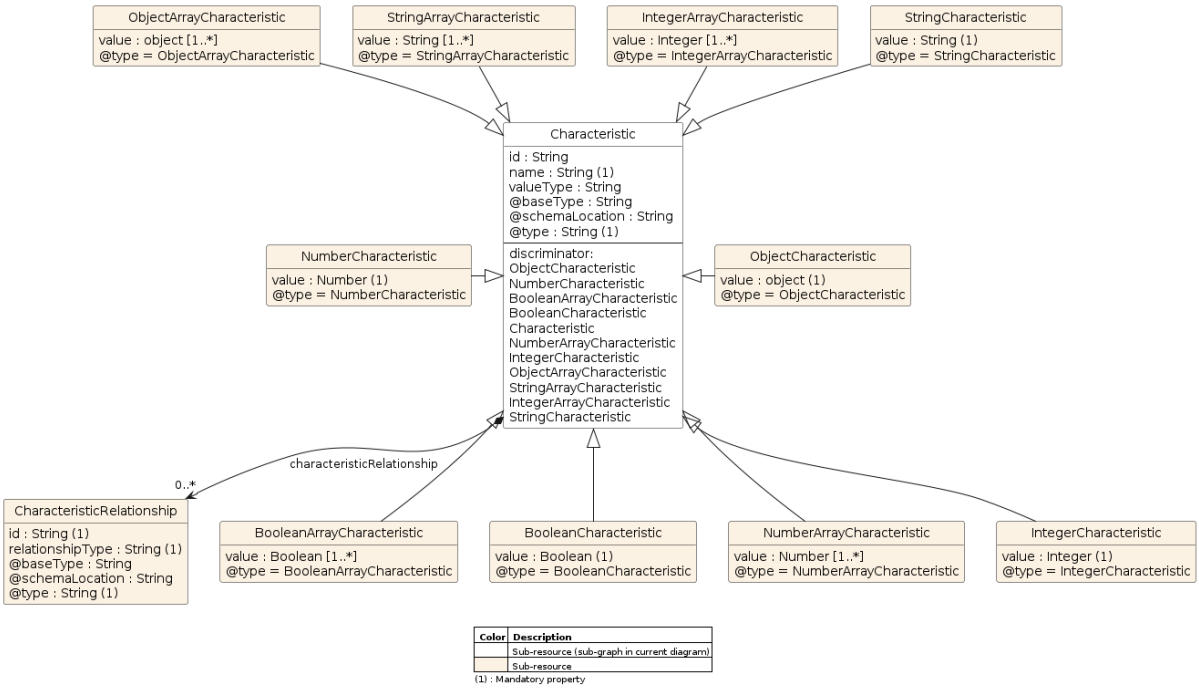


Figure 2 - Characteristic

Field descriptions

LogicalResource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined to represent further information indicating, for example, which services are not functional and the nature of the degradation. The operational state is 'enabled'; 'notInstalled': The resource represented by the managed object is not present, or is incomplete. The operational state is 'disabled'; 'logFull': This indicates a log full condition, which means that the maximum capacity of the log has been reached. ENUMERATED with values: * inTest * failed * powerOff * online * offline * offDuty * dependency * degraded * notInstalled * logFull

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.

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lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	LogicalResource can be instantiated as * ResourceFunction * SoftwareResource
autoModification	This property is present in subclasses
connectionPoint	This property is present in subclasses
connectivity	This property is present in subclasses
functionType	This property is present in subclasses
isDistributedCurrent	This property is present in subclasses
lastUpdate	This property is present in subclasses
priority	This property is present in subclasses
role	This property is present in subclasses
schedule	This property is present in subclasses
targetPlatform	This property is present in subclasses

Feature sub-resource fields

featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

place	A PlaceRef. Place reference.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.

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role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

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See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedResourceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the product. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRelationship sub-resource fields

relationshipType	A String. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRef. Resource reference, for when Resource is used by other entities.
resourceRelationshipCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [Characteristic] for details.

ConnectionPointRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Connection point version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceGraph sub-resource fields

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connection	A Connection. Resource graph edges.
description	A String. Description of the resource graph.
graphRelationship	A ResourceGraphRelationship. Relationships to other resource graphs.
id	A String. Unique identifier of the resource graph.
name	A String. Descriptive name for the resource graph.

Connection sub-resource fields

associationType	A String. Association type. ENUMERATED with values: * pointtoPoint * pointtoMultipoint
endpoint	An EndpointRef. Resource graph vertices connected by this edge.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Descriptive name for graph edge.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EndpointRef sub-resource fields

connectionPoint	A ConnectionPointRef. Connection point reference. A connection point provides a service access point (SAP) for input and/or output of the resource function.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isRoot	A Boolean. Directionality: true when endpoint is a source, false when a sink. If true for all endpoints connectivity is bidirectional. Default is true.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceGraphRelationship sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String. Semantic of the relationship. ENUMERATED with values: * adjacency * connectivity
resourceGraph	A ResourceGraphRef. Resource graph reference.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceGraphRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ScheduleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'LogicalResource' resource object.

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```
{
  "id": "839933ec-4c65-4fc2-987b-58ca5ba9cfab",
  "href": "https://host:port/tmf-
api/resourceInventoryManagement/v5/resource/839933ec-4c65-4fc2-987b-58ca5ba9cfab",
  "name": "MSISDN 491702283968",
  "description": "This  logical resource describes the MSISDN 491702283968",
  "startOperatingDate": "2025-09-23T00:00:00.000Z",
  "resourceVersion": "v1",
  "value": "491702283968",
  "administrativeState": "unlocked",
  "operationalState": "disabled",
  "usageState": "idle",
  "allocationStatus": "allocated",
  "resourceCharacteristic": [
    {
      "name": "MSISDN",
      "value": "491702283968",
      "valueType": "String",
      "@type": "StringCharacteristic"
    }
  ],
  "relatedParty": [
    {
      "role": "Owner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceSpecification": {
    "id": "ca344d7e-ea8a-49cd-9a5f-824970439c9e",
    "href": "https://host:port/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/ca344d7e-ea8a-49cd-9a5f-
824970439c9e",
    "@type": "ResourceSpecificationRef",
    "@referredType": "LogicalResourceSpecification"
  },
  "@type": "LogicalResource",
  "@baseType": "Resource"
}
```


PhysicalResource resource

Physical resource is a type of resource that describes the common set of attributes shared by all concrete physical resources (e.g. EQUIPMENT) in the inventory.

Resource model

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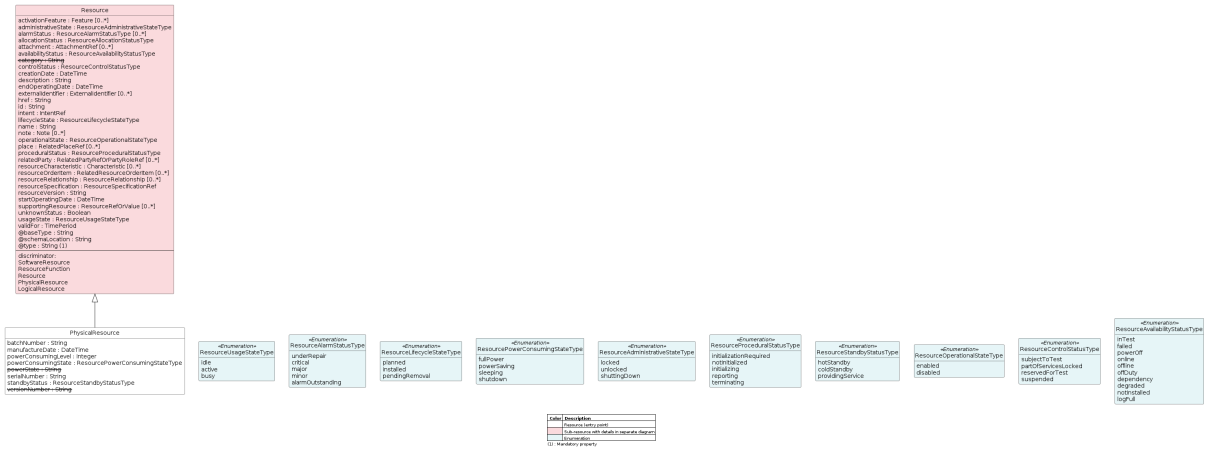


Figure 3 - PhysicalResource

See Figure 2 for Characteristic diagrams.

Field descriptions

PhysicalResource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined to represent further information indicating, for example, which services are not functional and the nature of the degradation. The operational state is 'enabled'; 'notInstalled': The resource represented by the managed object is not present, or is incomplete. The operational state is 'disabled'; 'logFull': This indicates a log full condition, which means that the maximum capacity of the log has been reached. ENUMERATED with values: * inTest * failed * powerOff * online * offline * offDuty * dependency * degraded * notInstalled * logFull

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batchNumber	A String. A unique identifier assigned to a specific group of physical resources produced or received under the same conditions for tracking and traceability in the supply chain.
~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.

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lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the ISO 8601 date time standard format. This is an optional attribute.
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
powerConsumingLevel	An Integer. The power consuming level attribute (defined by ITU-T M.3701) indicates the current percentage of power consuming of the resource represented by the managed object, where 100 percent represents the maximum power consumption of the device. It is single valued and read-only by management functions.

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powerConsumingState	A ResourcePowerConsumingStateType. Value enumeration for the 'resourcePowerConsumingState' of a Resource; the power consuming state attribute (defined by ITU-T M.3701) indicates the power consuming policy of the resource represented by the managed object, it is single valued and read-write by management functions.; defined values: 'fullPower': The resource runs in full power and will not consider power saving. If a large device contains parts that also support this attribute, when the container object set to full power state, all its parts shall turn into full power state; 'powerSaving': The resource is (or may be) partially running to save power; depending on how much the payload is lower than the maximum capacity of the resource, the degree of saving will be automatically decided by the resource. If power saving is not supported by the resource, attempting to change to this state will have the same effect as full power state. If a large device contains parts that also support this attribute, when the container object is set to power saving state, its parts may still remain in full power state. But if one of the parts turns into power saving state, the container object shall turn into power saving state; 'sleeping': The resource is not running but keeps a snapshot of its memory before sleeping, and can resume running mode quickly. In this mode, the power is only consumed to meet the need of maintaining the memory. If it is not supported by the resource, attempting to change to this state will result in a failure; 'shutdown': The resource has been powered off and does not consume any more power; if it is not supported by the resource, attempting to change to this state will result in a failure. ENUMERATED with values: * fullPower * powerSaving * sleeping * shutdown
~~powerState~~	DEPRECATED A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off; NOTE: This attribute is replaced by the 'powerConsumingState' attribute, to align with the official name as defined by ITU-T.

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and PartNumber attributes are used to identify different types of hardware items.
standbyStatus	A ResourceStandbyStatusType. Value enumeration for the 'resourceStandbyStatus' of a Resource; the standby status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the working mode for standby objects, i.e. whether it is providing service or standby as a back-up resource; it is single valued and read-only by management functions; defined values: 'hotStandby': The resource is not providing service, but is operating in synchronism with another resource that is to be backed-up (e.g., a computer shadowing another computer). A resource with a hot standby status will be immediately able to take over the role of the resource to be backed up, without the need for initialization activity, and will contain the same information as the resource to be backed up; 'coldStandby': The resource is to back up another resource, but is not synchronized with that resource. A resource with a cold standby status will not be immediately able to take over the role of a resource to be backed up, and will require some initialization activity; 'providingService': The back up resource is providing service and is backing up another resource. ENUMERATED with values: * hotStandby * coldStandby * providingService
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).

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unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
~~versionNumber~~	DEPRECATED A String. This is a string that identifies the version of this physical resource. This is an optional attribute. NOTE that this attribute is deprecated as it overlaps with the Resource.resourceVersion attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

See [Feature] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

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IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

See [FeatureRelationship] for details.

PolicyRef sub-resource fields

See [PolicyRef] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

See [IntentRef] for details.

Note sub-resource fields

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See [Note] for details.

RelatedPlaceRef sub-resource fields

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

See [PlaceRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

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ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedResourceOrderItem sub-resource fields

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

See [ResourceSpecificationRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'PhysicalResource' resource object.


```
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "enabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
```


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```
      "partyOrPartyRole": {
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
```


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```
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```


Resource resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

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Resource model

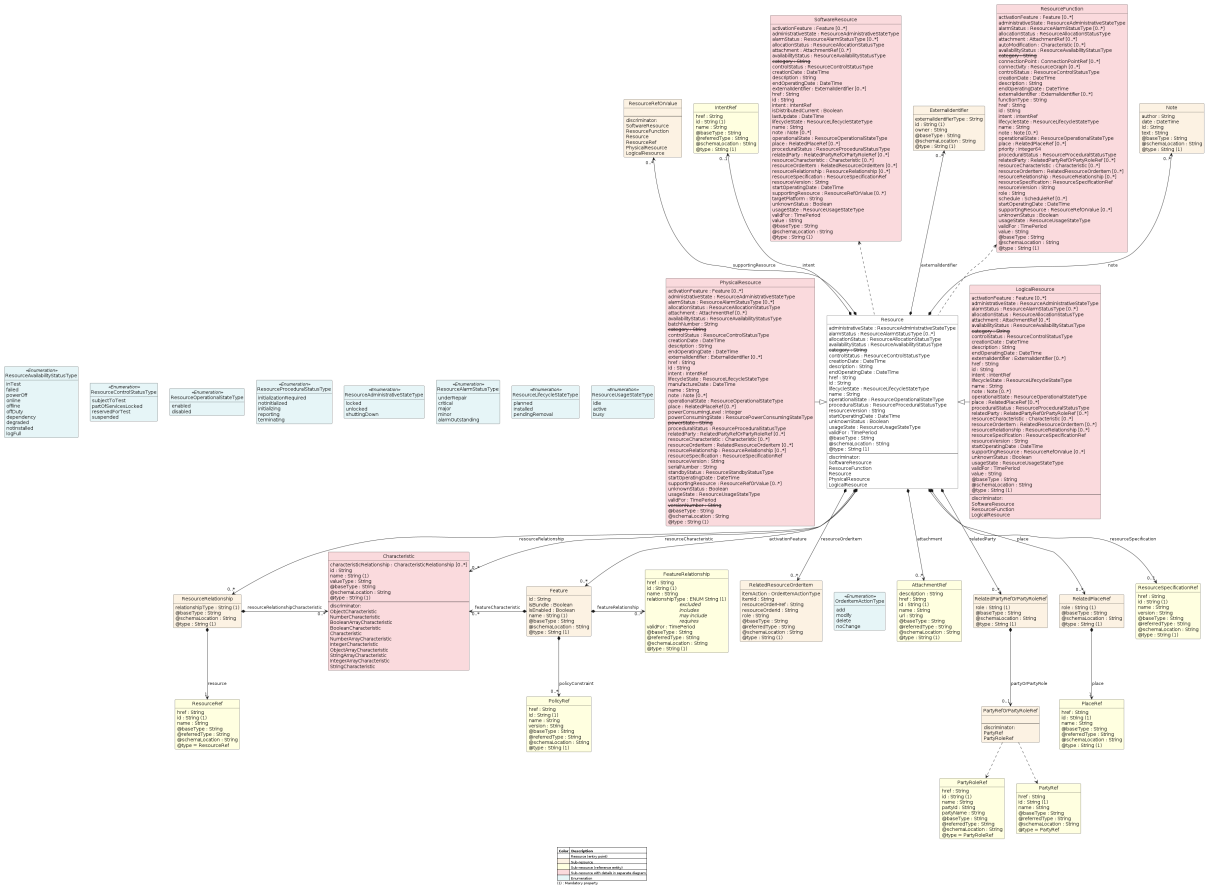


Figure 4 - Resource

See Figure 2 for Characteristic diagrams.

Field descriptions

Resource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown
alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined to represent further information indicating, for example, which services are not functional and the nature of the degradation. The operational state is 'enabled'; 'notInstalled': The resource represented by the managed object is not present, or is incomplete. The operational state is 'disabled'; 'logFull': This indicates a log full condition, which means that the maximum capacity of the log has been reached. ENUMERATED with values: * inTest * failed * powerOff * online * offline * offDuty * dependency * degraded * notInstalled * logFull

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.

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lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	Resource can be instantiated as * LogicalResource * PhysicalResource
batchNumber	This property is present in subclasses
manufactureDate	This property is present in subclasses
powerConsumingLevel	This property is present in subclasses
powerConsumingState	This property is present in subclasses
powerState	This property is present in subclasses
serialNumber	This property is present in subclasses
standbyStatus	This property is present in subclasses
value	This property is present in subclasses
versionNumber	This property is present in subclasses

Feature sub-resource fields

See [Feature] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

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ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

See [FeatureRelationship] for details.

PolicyRef sub-resource fields

See [PolicyRef] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

See [IntentRef] for details.

Note sub-resource fields

See [Note] for details.

RelatedPlaceRef sub-resource fields

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

See [PlaceRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

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See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedResourceOrderItem sub-resource fields

See [RelatedResourceOrderItem] for details.

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ResourceRelationship sub-resource fields

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

See [ResourceSpecificationRef] for details.

ResourceFunction resource

A ResourceFunction is a behavior to transform inputs of any nature into outputs of any nature independently from the way it is provided.

Resource model

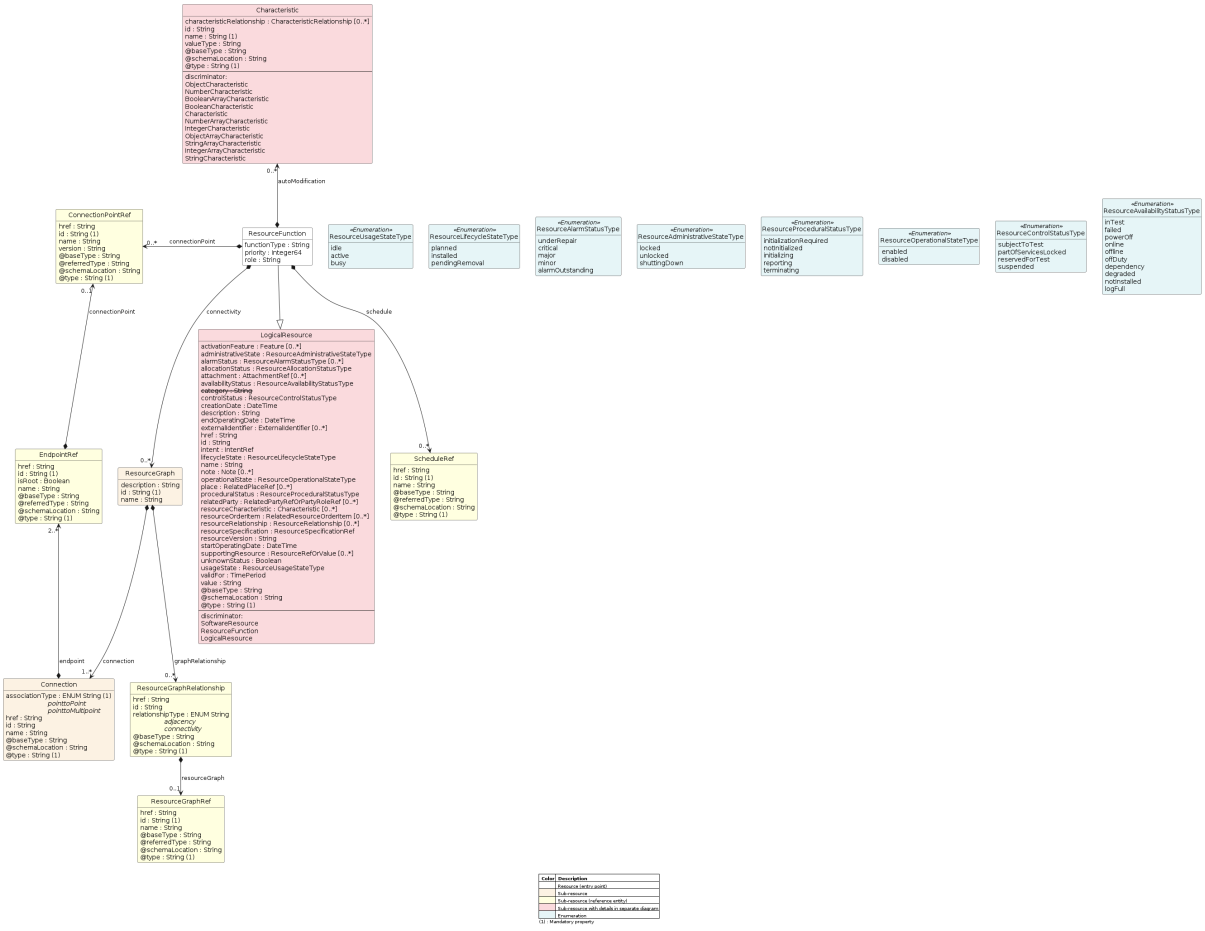


Figure 5 - ResourceFunction

See Figure 2 for Characteristic diagrams.

Field descriptions

ResourceFunction fields

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activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown
alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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autoModification	A Characteristic. List of the kinds of auto-modifications that are applied to a given network service e.g what can be scaled. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined to represent further information indicating, for example, which services are not functional and the nature of the degradation. The operational state is 'enabled'; 'notInstalled': The resource represented by the managed object is not present, or is incomplete. The operational state is 'disabled'; 'logFull': This indicates a log full condition, which means that the maximum capacity of the log has been reached. ENUMERATED with values: * inTest * failed * powerOff * online * offline * offDuty * dependency * degraded * notInstalled * logFull

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
connectionPoint	A ConnectionPointRef. External connection points of the resource function. These are the service access points (SAP) where inputs and outputs of the function are available.
connectivity	A ResourceGraph. Internal connectivity of contained resource functions.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
functionType	A String. A type of the Resource Function as specified by the provider of the API.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
priority	An Integer64. Priority of the Resource Function. Decides what happens in a contention scenario.

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
role	A String. Role of the Resource Function. Used when Resource Function is a component of a composite Resource Function and the exact role of the service within the composite is not clear from descriptor/location.
schedule	A ScheduleRef. This is a reference to a schedule. Allows consumers to schedule modifications to the service at certain times.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

See [Feature] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

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StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

See [FeatureRelationship] for details.

PolicyRef sub-resource fields

See [PolicyRef] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

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See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

ConnectionPointRef sub-resource fields

See [ConnectionPointRef] for details.

ResourceGraph sub-resource fields

See [ResourceGraph] for details.

Connection sub-resource fields

See [Connection] for details.

EndpointRef sub-resource fields

See [EndpointRef] for details.

ResourceGraphRelationship sub-resource fields

See [ResourceGraphRelationship] for details.

ResourceGraphRef sub-resource fields

See [ResourceGraphRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

See [IntentRef] for details.

Note sub-resource fields

See [Note] for details.

RelatedPlaceRef sub-resource fields

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

See [PlaceRef] for details.

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RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

RelatedResourceOrderItem sub-resource fields

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

See [ResourceSpecificationRef] for details.

ScheduleRef sub-resource fields

See [ScheduleRef] for details.

SoftwareResource resource

Abstract class describing the common set of attributes shared by all concrete software resources (e.g. API, InstalledSoftware).

Resource model

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Figure 6 - SoftwareResource

Field descriptions

SoftwareResource fields

activationFeature	A Feature. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined to represent further information indicating, for example, which services are not functional and the nature of the degradation. The operational state is 'enabled'; 'notInstalled': The resource represented by the managed object is not present, or is incomplete. The operational state is 'disabled'; 'logFull': This indicates a log full condition, which means that the maximum capacity of the log has been reached. ENUMERATED with values: * inTest * failed * powerOff * online * offline * offDuty * dependency * degraded * notInstalled * logFull

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
isDistributedCurrent	A Boolean. True defines this SoftwareResource as being currently distributed across several Resource nodes.
lastUpdate	A DateTime. The time of the last update of this resource.

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lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship. A list of relationships between this resource and other resources, differentiated by relationshipType.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue. A collection of resources that support this resource (bundling, link ResourceSpecification).
targetPlatform	A String. Defines the target platform that the SoftwareResource will be deployed on.
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

See [Feature] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

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FeatureRelationship sub-resource fields

See [FeatureRelationship] for details.

PolicyRef sub-resource fields

See [PolicyRef] for details.

AttachmentRef sub-resource fields

See [AttachmentRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

See [IntentRef] for details.

Note sub-resource fields

See [Note] for details.

RelatedPlaceRef sub-resource fields

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

See [PlaceRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

Characteristic sub-resource fields

See [Characteristic] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

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See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

RelatedResourceOrderItem sub-resource fields

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

See [ResourceSpecificationRef] for details.

Notification Resource Models

4 notifications are defined for this API.

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Notifications related to Resource:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

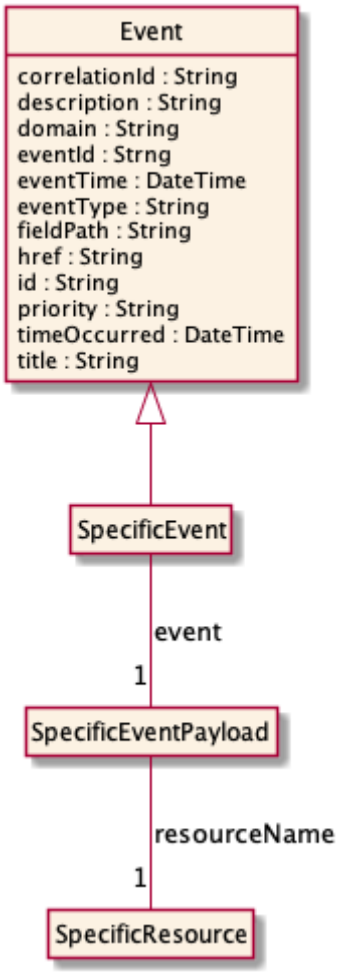


Figure 7 Notification Pattern

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on LogicalResource

See operations on Resource.

Operations on PhysicalResource

See operations on Resource.

Operations on Resource

List or find Resource objects

`GET /resource?fields=…`

Description

This operation list Resource entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of Resource(s) fulfilling the provided query criteria. The given criteria is the name (Router) and allocationStatus (allocated).

Request

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```
GET /resource?filter=name=Router&allocationStatus=allocated
Content-Type: application/json
```

Response

```
200
[
  {
    "id": "ca239a70-5810-433f-a251-ba5277a5af72",
    "href": "http://host:port/resourceInventoryManagement/v5/resource/80ca239a70-
5810-433f-a251-ba5277a5af7244",
    "name": "Router",
    "description": "This is a Router resource.",
    "startOperatingDate": "2025-05-13T00:00:00.000Z",
    "operationalState": "enabled",
    "usageState": "idle",
    "administrativeState": "unlocked",
    "lifecycleState": "installed",
    "availabilityStatus": "online",
    "controlStatus": "subjectToTest",
    "standbyStatus": "providingService",
    "unknownStatus": false,
    "powerConsumingState": "fullPower",
    "powerConsumingLevel": 100,
    "allocationStatus": "allocated",
    "resourceVersion": "v2",
    "relatedParty": [
      {
        "role": "user",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
          "id": "fcba712d-0462-40e5-8d81-6a3439980278",
          "name": "John Doe",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    ],
    "note": [
      {
        "text": "Router installed at customer's premises, available for testing
purposes.",
        "date": "2025-05-13T00:00:00.000Z",
        "@type": "Note"
      }
    ],
    "place": [
      {
        "role": "installationAddress",
        "@type": "RelatedPlaceRef",
        "place": {
          "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
          "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
          "@type": "PlaceRef",
          "@referredType": "GeographicAddress"
```

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```
        }
      }
    ],
    "resourceRelationship": [
      {
        "relationshipType": "contains",
        "@type": "ResourceRelationship",
        "resource": {
          "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
          "href":
"http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-6ac3-4e96-ac8d-
c32343a05b2d",
          "@type": "ResourceRef",
          "@referredType": "PhysicalResource"
        }
      },
      {
        "relationshipType": "contains",
        "@type": "ResourceRelationship",
        "resource": {
          "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
          "href":
"http://host:port/resourceInventoryManagement/v5/resource/00f00279-ef09-4592-bcbf-
7ced615903d2",
          "@type": "ResourceRef",
          "@referredType": "PhysicalResource"
        }
      }
    ],
    "resourceSpecification": {
      "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
      "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
      "@type": "ResourceSpecificationRef",
      "@referredType": "PhysicalResourceSpecification"
    },
    "externalIdentifier": [
      {
        "externalIdentifierType": "Supplier-Router-Id",
        "id": "B-AGB-6414-X",
        "owner": "Supplier-B",
        "@type": "ExternalIdentifier"
      }
    ],
    "supportingResource": [
      {
        "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    ],
    "resourceCharacteristic": [
      {
        "name": "casing size",
        "valueType": "Quantity",
        "value": {
          "amount": 19,
          "units": "\""
        },
        "@type": "ObjectCharacteristic"
```


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```
      }
    ],
    "resourceOrderItem": [
      {
        "role": "supplier purchase",
        "itemAction": "add",
        "itemId": "1",
        "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
        "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
        "@type": "RelatedResourceOrderItem",
        "@referredType": "ResourceOrderItem"
      }
    ],
    "attachment": [
      {
        "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
        "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-
4204-9efd-7089a06aa89a",
        "@type": "AttachmentRef",
        "@referredType": "Attachment"
      }
    ],
    "@type": "PhysicalResource",
    "@baseType": "Resource"
  }
]
```


Retrieves a Resource by ID

`GET /resource/{id}?fields=…`

Description

This operation retrieves a Resource entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific Resource. The given criteria is the Resource id 839933ec-4c65-4fc2-987b-58ca5ba9cfab.

Request

```
GET /resource/839933ec-4c65-4fc2-987b-58ca5ba9cfab
Content-Type: application/json
```

Response


```
200
{
  "id": "839933ec-4c65-4fc2-987b-58ca5ba9cfab",
  "href": "https://host:port/tmf-
```


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```
api/resourceInventoryManagement/v5/resource/839933ec-4c65-4fc2-987b-58ca5ba9cfab",
  "name": "MSISDN 491702283968",
  "description": "This  logical resource describes the MSISDN 491702283968",
  "startOperatingDate": "2025-09-23T00:00:00.000Z",
  "resourceVersion": "v1",
  "value": "491702283968",
  "administrativeState": "unlocked",
  "operationalState": "disabled",
  "usageState": "idle",
  "allocationStatus": "allocated",
  "resourceCharacteristic": [
    {
      "name": "MSISDN",
      "value": "491702283968",
      "valueType": "String",
      "@type": "StringCharacteristic"
    }
  ],
  "relatedParty": [
    {
      "role": "Owner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "resourceSpecification": {
    "id": "ca344d7e-ea8a-49cd-9a5f-824970439c9e",
    "href": "https://host:port/tmf-
api/resourceCatalogManagement/v5/resourceSpecification/ca344d7e-ea8a-49cd-9a5f-
824970439c9e",
    "@type": "ResourceSpecificationRef",
    "@referredType": "LogicalResourceSpecification"
  },
  "@type": "LogicalResource",
  "@baseType": "Resource"
}
```


Creates a Resource

`POST /resource?fields=…`

Description

This operation creates a Resource entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	M

Usage samples

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Creation of a new Resource (a Router) containing Sub-Resources (two Ports) via a POST operation, assuming the 'Port' Resources (ids 0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d and 35e126c3-def2-4e6d-8e0d-defb1c5dc5c9) have already been created before.

Request

```
POST /resource
Content-Type: application/json
{
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "enabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
```

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```
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
```


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```
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```


Response


```
201
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/80ca239a70-5810-
433f-a251-ba5277a5af7244",
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "enabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
```


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```
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
```


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```
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```


Updates partially a Resource

`PATCH /resource/{id}?fields=…`

Description

This operation allows partial updates of a Resource entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

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Non Patchable Attributes	Rule
activationFeature	
administrativeState	
alarmStatus	
allocationStatus	
attachment	
availabilityStatus	
category	
controlStatus	
creationDate	
description	
endOperatingDate	
externalIdentifier	
href	
id	
intent	
lifecycleState	
name	
note	
operationalState	
place	
proceduralStatus	
relatedParty	
resourceCharacteristic	
resourceOrderItem	
resourceRelationship	
resourceSpecification	
resourceVersion	
startOperatingDate	
supportingResource	
unknownStatus	
usageState	
validFor	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable

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Non Patchable Attributes	Rule
@type	@type is immutable

Patchable Attributes	Rule

Usage samples

Update a Resource using json-patch-query+json

Request

```
PATCH /resource/{id}
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": ".note[?(@.date=='2025-05-13T00:00:00.000Z')]",
    "value": {
      "text": "Port 1 configured"
    }
  }
]
```

Response

```
200
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "enabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
```

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```
      }
    }
  ],
  "note": [
    {
      "text": "Port 1 configured",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
```


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```
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```


Update a Resource using json-patch

Request

```
PATCH /resource/{id}
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/administrativeState",
    "value": "locked"
  },
  {
```

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```
    "op": "replace",
    "path": "/description",
    "value": "This is the Router owned by John Doe."
  },
  {
    "op": "add",
    "path": "/resourceRelationship",
    "value": {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "3df76e2c-211f-4aea-9843-027ef42449f6",
        "name": "Port",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  }
]
```

Response

```
200
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "name": "Router",
  "description": "This is the Router owned by John Doe.",
  "startOperatingDate": "2025-10-21T00:00:00.000Z",
  "operationalState": "enabled",
  "usageState": "idle",
  "administrativeState": "locked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
```

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```
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "id": "f01a6e7a-a9e7-4427-86aa-3bcfc9e44a4d",
      "resource": {
        "id": "3df76e2c-211f-4aea-9843-027ef42449f6",
        "name": "Port",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
```


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```
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```


PATCH Resource request leveraging merge-patch+json application example

Request

```
PATCH /resource/{id}
Content-Type: application/merge-patch+json
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
```

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```
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "@type": "PhysicalResource",
  "operationalState": "disabled"
}
```


Response


```
200
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "disabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
```


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```
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
```


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```
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
}
```

PATCH Resource request leveraging json application example

Request

```
PATCH /resource/{id}
Content-Type: application/json
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "@type": "PhysicalResource",
  "operationalState": "disabled"
}
```

Response

```
200
{
  "id": "ca239a70-5810-433f-a251-ba5277a5af72",
  "href": "http://host:port/resourceInventoryManagement/v5/resource/ca239a70-5810-
433f-a251-ba5277a5af72",
  "name": "Router",
  "description": "This is a Router resource.",
  "startOperatingDate": "2025-05-13T00:00:00.000Z",
  "operationalState": "disabled",
  "usageState": "idle",
  "administrativeState": "unlocked",
  "lifecycleState": "installed",
  "availabilityStatus": "online",
  "controlStatus": "subjectToTest",
```

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```
  "standbyStatus": "providingService",
  "unknownStatus": false,
  "powerConsumingState": "fullPower",
  "powerConsumingLevel": 100,
  "allocationStatus": "allocated",
  "resourceVersion": "v2",
  "relatedParty": [
    {
      "role": "user",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "fcba712d-0462-40e5-8d81-6a3439980278",
        "href": "https://host:port/tmf-api/partyManagement/v5/individual/fcba712d-
0462-40e5-8d81-6a3439980278",
        "name": "John Doe",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    }
  ],
  "note": [
    {
      "text": "Router installed at customer's premises, available for testing
purposes.",
      "date": "2025-05-13T00:00:00.000Z",
      "@type": "Note"
    }
  ],
  "place": [
    {
      "role": "installationAddress",
      "@type": "RelatedPlaceRef",
      "place": {
        "id": "859ad898-2308-4879-a09c-33a35cbd1c44",
        "href":
"https://host:port/geographicAddressManagement/v5/geographicAddress/859ad898-2308-
4879-a09c-33a35cbd1c44",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress"
      }
    }
  ],
  "resourceRelationship": [
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "0d5c8d46-6ac3-4e96-ac8d-c32343a05b2d",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/0d5c8d46-
6ac3-4e96-ac8d-c32343a05b2d",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
      }
    },
    {
      "relationshipType": "contains",
      "@type": "ResourceRelationship",
      "resource": {
        "id": "00f00279-ef09-4592-bcbf-7ced615903d2",
        "href": "http://host:port/resourceInventoryManagement/v5/resource/00f00279-
ef09-4592-bcbf-7ced615903d2",
        "@type": "ResourceRef",
        "@referredType": "PhysicalResource"
```


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```
      }
    }
  ],
  "resourceSpecification": {
    "id": "74ed54b9-008f-4d1d-87f0-b79a283c6ce5",
    "href": "
http://host:port/resourceCatalogManagement/v5/resourceSpecification/74ed54b9-008f-
4d1d-87f0-b79a283c6ce5",
    "@type": "ResourceSpecificationRef",
    "@referredType": "PhysicalResourceSpecification"
  },
  "externalIdentifier": [
    {
      "externalIdentifierType": "Supplier-Router-Id",
      "id": "B-AGB-6414-X",
      "owner": "Supplier-B",
      "@type": "ExternalIdentifier"
    }
  ],
  "supportingResource": [
    {
      "id": "e451a1f1-f9c9-4040-a4eb-97a71c2ddc53",
      "href": "http://host:port/resourceInventoryManagement/v5/resource/e451a1f1-
f9c9-4040-a4eb-97a71c2ddc53",
      "@type": "ResourceRef",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceCharacteristic": [
    {
      "name": "casing size",
      "valueType": "Quantity",
      "value": {
        "amount": 19,
        "units": "\""
      },
      "@type": "ObjectCharacteristic"
    }
  ],
  "resourceOrderItem": [
    {
      "role": "supplier purchase",
      "itemAction": "add",
      "itemId": "1",
      "resourceOrderId": "10720d6f-e5a6-4590-a3c7-89194e3039f3",
      "resourceOrderHref":
"http://host:port/resourceOrderManagement/v5/resourceOrder/10720d6f-e5a6-4590-a3c7-
89194e3039f3",
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem"
    }
  ],
  "attachment": [
    {
      "id": "439ad8f9-bcf9-4204-9efd-7089a06aa89a",
      "href": "http://host:port/documentManagement/v5/attachment/439ad8f9-bcf9-4204-
9efd-7089a06aa89a",
      "@type": "AttachmentRef",
      "@referredType": "Attachment"
    }
  ],
  "@type": "PhysicalResource",
  "@baseType": "Resource"
```


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```
}
```

Deletes a Resource

`DELETE /resource/{id}`

Description

This operation deletes a Resource entity.

Operations on ResourceFunction

See operations on LogicalResource, Resource.

Operations on SoftwareResource

See operations on LogicalResource, Resource.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description

Contributors to Document

Version Number	Date	Modified by	Description
5.0.0	04.06.2024 - 05.02.2026	Lutz Bettge Alexander Klörig Deutsche Telekom	migrated to version 5
5.0.0	05.02.2026	Amani Dkhil Sofrecom Tunisie	review of Conformance Profile
5.0.0	05.02.2026	Jonathan Goldberg TM Forum	overall review

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